

Movie Recommendation System using Big Data Analytics

Midterm Evaluation Report

Subject: Big Data

Submitted by

Krishanu Anand (23CSU166)
Yathansh Jhakar (23CSU352)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING,

SCHOOL OF ENGINEERING AND TECHNOLOGY

THE NORTHCAP UNIVERSITY

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Abstract

This project builds an end-to-end Big Data pipeline on the MovieLens ml-25m dataset (25,000,095 ratings from 162,541 users on 59,047 movies). Data is ingested into Google Cloud Storage and processed with PySpark on Google Cloud Dataproc using a Bronze → Silver → Gold medallion architecture. Spark SQL and window functions produce analysis-ready tables, and a Spark MLlib ALS collaborative-filtering model predicts ratings with a test RMSE of 0.7874 (vs. 1.0578 for a global-mean baseline) and generates a personalised Top-10 list for every user. Results are presented in an interactive Streamlit dashboard, and the workflow is orchestrated with a runner script and an Airflow DAG.

Source code: https://github.com/anand-krishanu/AttractorField

1. Problem Definition

Streaming catalogues contain tens of thousands of titles; users struggle to discover movies they will enjoy, and content teams lack visibility into how tastes evolve. Manually curated lists do not scale to millions of users.

Objectives

- Ingest and store a large real-world ratings dataset in the cloud with full lineage metadata.
- Clean and validate the data (nulls, duplicates, invalid values, orphan keys, inconsistent text).
- Produce analysis-ready Gold tables: popularity, weighted ratings, genre trends, user segments, KPIs.
- Train and evaluate a scalable recommender (ALS) that beats a naive baseline.
- Deliver personalised Top-10 recommendations and an interactive dashboard built on real outputs.

Target users & expected outcomes

- Viewers: personalised, unseen movie recommendations and 'similar movies'.
- Product/content teams: KPIs, genre trends and user segments to guide acquisition and promotion.
- Outcome: an automated, reproducible pipeline from raw data source to end-user dashboard.

2. Dataset

MovieLens 25M is a stable benchmark dataset released by GroupLens Research (University of Minnesota) in 2019. It contains explicit 0.5–5 star ratings collected on movielens.org between 1995 and 2019; each user has rated at least 20 movies. Files: ratings.csv (userId, movieId, rating, timestamp), movies.csv (title with year, pipe-separated genres), tags.csv (free-text user tags) and links.csv (IMDb/TMDb ids). Source: https://grouplens.org/datasets/movielens/25m/ . The smaller ml-latest-small release (100k ratings) was used for local development and testing.

Citation: F. Maxwell Harper and Joseph A. Konstan. 2015. The MovieLens Datasets: History and Context. ACM Transactions on Interactive Intelligent Systems 5, 4, Article 19. https://doi.org/10.1145/2827872

3. Architecture & Technologies

GroupLens → ingest.py → GCS raw/ → Bronze (Parquet) → Silver (clean, partitioned) → Gold (analytics) + MLlib ALS (output/) → Streamlit dashboard. Orchestration: run_pipeline.py (local Docker), scripts/dataproc_run.ps1 (Google Cloud) and an Airflow DAG (dags/movie_reco_dag.py).

Layer	Technology
Storage	Google Cloud Storage (data lake), Parquet (columnar, partitioned)
Processing	Apache Spark 3.5 / PySpark, Spark SQL, window functions
Machine learning	Spark MLlib ALS, RegressionEvaluator, RankingEvaluator
Cloud compute	Google Cloud Dataproc 2.2 (1 × e2-standard-2 master, 2 × e2-highmem-4 workers)
Orchestration	Apache Airflow DAG (Dataproc operators), PowerShell/Python runners
Visualisation	Streamlit + Plotly dashboard
Dev environment	Docker image apache/spark 3.5.3 (Java 17) mirroring Dataproc; GitHub with PR workflow

4. Methodology

4.1 Ingestion & Bronze layer

ingest.py downloads the official zip, writes a manifest (source URL, timestamp, file sizes) and uploads the raw CSVs to GCS — on Dataproc this runs on the cluster so data never passes through the client. bronze.py reads the CSVs with explicit schemas (no inference) and adds ingest_ts, source_file and dataset_version columns before writing Parquet, preserving the raw values for lineage and reprocessing.

4.2 Silver layer – cleaning

silver.py removes null keys, out-of-range ratings and invalid timestamps; resolves duplicate (user, movie) ratings by keeping the latest one with a row_number() window; drops ratings that reference unknown movies (left-semi join); converts epoch seconds to timestamps; extracts the release year from the title with a regex; turns the pipe-separated genres into arrays (mapping '(no genres listed)' to null); lower-cases and de-duplicates tags; and joins IMDb/TMDb links. Ratings are written partitioned by rating_year. Each check is logged:

Table	Check	Rows before	Rows after	Removed
movies	null id/title	62,423	62,423	0
movies	duplicate movieId	62,423	62,423	0
ratings	null fields	25,000,095	25,000,095	0
ratings	invalid rating/timestamp	25,000,095	25,000,095	0
ratings	duplicate user-movie	25,000,095	25,000,095	0
ratings	orphan movieId	25,000,095	25,000,095	0
tags	null/empty tag	1,093,360	1,093,360	0
tags	duplicate tag	1,093,360	1,093,359	1

4.3 Gold layer – analytics

- movie_stats: count, mean, std-dev and an IMDb-style Bayesian weighted rating WR = v/(v+m)·R + m/(v+m)·C (m = 90th percentile of vote counts) so that low-count movies do not dominate.
- top_movies_by_genre: rank() over a window partitioned by genre (top 20 per genre).
- genre_trends: Spark SQL over a temp view with LAG() to compute year-over-year growth per genre.
- user_activity: per-user volume, average rating, first/last activity and a segment (casual→power).
- user_top_rated, rating_distribution, top_tags and yearly_kpis (with a running cumulative total).

4.4 Recommendation model – Spark MLlib ALS

Alternating Least Squares factorises the sparse user × movie rating matrix into low-rank user and item factor matrices; predicted rating = dot product of the two factor vectors. A first run on all movies showed that titles with only 2–10 ratings received extreme factors and flooded every user's top-10 (precision@10 = 0), so the model is trained only on movies with at least 50 ratings (13,176 movies, still the vast majority of all ratings). Data was split 70/10/20 (train/validation/test, seed 42). A grid over rank ∈ {10, 20} and regParam ∈ {0.05, 0.1} (maxIter 10, non-negative factors, coldStartStrategy=drop) was evaluated on the validation split; the best configuration was refit on train+val and evaluated once on the held-out test split. Ranking quality was measured with precision@10 and NDCG@10 on a sample of 2,000 users, treating test movies rated ≥ 4 as relevant. Recommendations exclude movies the user has already rated. Item-item similarity was computed as cosine similarity between ALS item factors using Spark SQL higher-order functions.

5. Results & Analysis

5.1 Exploratory analytics

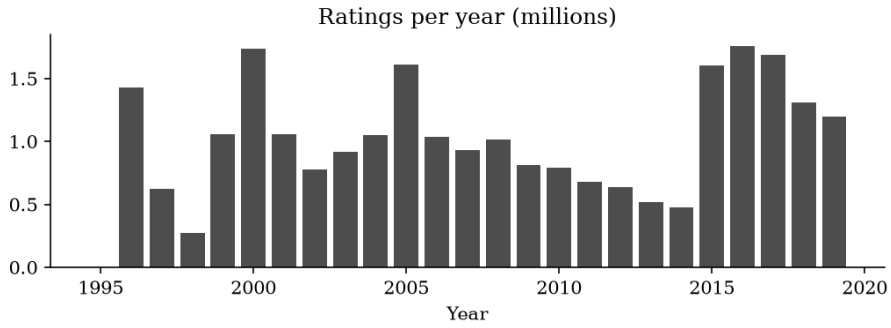


Figure 1: Rating volume per year

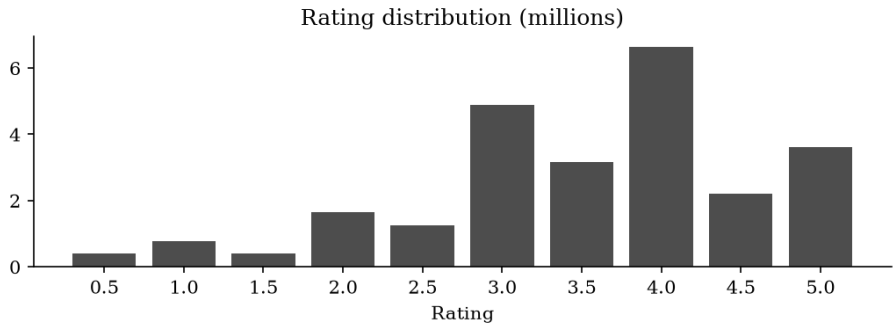


Figure 2: Distribution of ratings

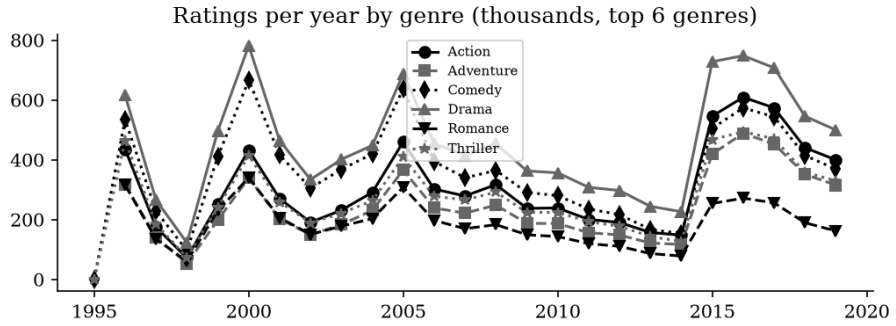


Figure 3: Genre popularity over time

Ratings are skewed positive: 50% of ratings are 4 stars or more and whole-star ratings dominate half-stars. The busiest year was 2016. Users rated on average 154 movies (median 71), showing a long-tail activity distribution; 6.0% of users are 'power' users with 500+ ratings.

Top-rated movies (Drama, by weighted rating)

Rank	Title	Year	Ratings	Avg rating	Weighted rating
1	Shawshank Redemption, The	1994	81,482	4.414	4.410
2	Godfather, The	1972	52,498	4.324	4.319
3	Godfather: Part II, The	1974	34,188	4.262	4.255
4	Schindler's List	1993	60,411	4.248	4.244
5	Seven Samurai (Shichinin no samurai)	1954	13,367	4.255	4.237

5.2 Model evaluation

Rank	regParam	validation RMSE
10	0.050	0.800
10	0.100	0.814
20	0.050	0.791
20	0.100	0.811

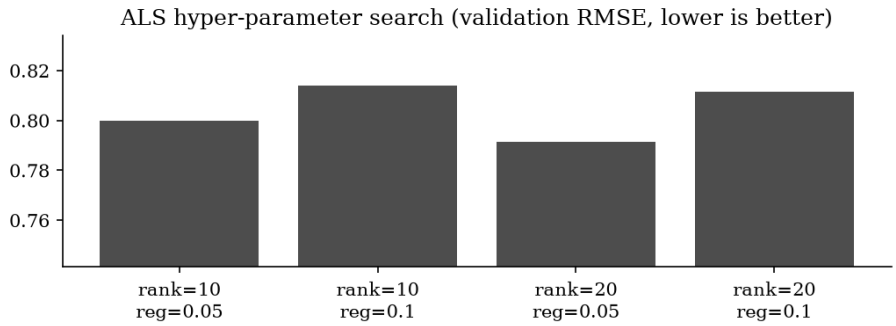


Figure 4: ALS hyper-parameter search

Metric	Value
Best parameters	rank=20, regParam=0.05
Train / Val / Test rows	17,252,352 / 2,462,699 / 4,929,877
Test RMSE (ALS)	0.787
Test RMSE (global-mean baseline)	1.058
Test MAE	0.609
Precision@10	0.024
NDCG@10	0.031

ALS reduces RMSE by 25.6% relative to always predicting the global mean, i.e. on average its predictions are within ~0.61 stars of the true rating. Ranking metrics are low in absolute terms, which is expected for top-10 retrieval from tens of thousands of movies when only held-out test ratings count as relevant; they serve as a baseline for future improvements.

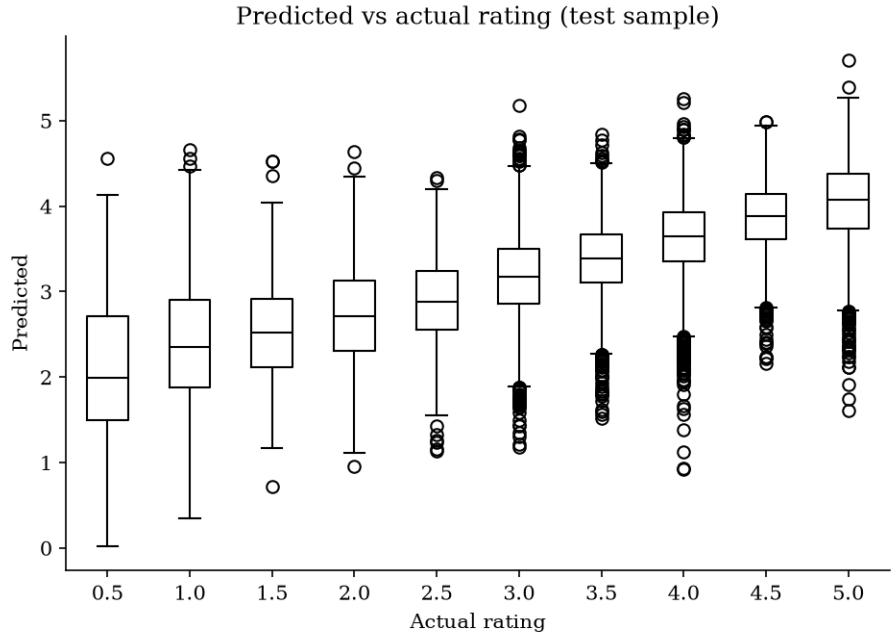


Figure 5: Predicted vs actual ratings on the test split

5.3 Sample recommendations

User 148 – favourite movies (highest rated):

Title	Rating
Paths of Glory	5.000

Good, the Bad and the Ugly, The (Buono, il brutto, il cattivo, Il)	5.000
Chinatown	5.000
Life Is Beautiful (La Vita è bella)	5.000
Seven Samurai (Shichinin no samurai)	5.000

User 148 – Top-10 recommendations (unseen):

Rank	Title	Year	Predicted rating
1	Planet Earth II	2016	4.540
2	Planet Earth	2006	4.533
3	Band of Brothers	2001	4.487
4	In the blue sea, in the white foam.	1984	4.475
5	Life	2009	4.437
6	Cosmos	–	4.426
7	The Adventures of Sherlock Holmes and Dr. Watson: Bloody Signature	1979	4.422
8	Alone in the Wilderness	2004	4.403
9	Blue Planet II	2017	4.310
10	Adventures Of Sherlock Holmes And Dr. Watson: The Twentieth Century Approaches	1986	4.298

6. Dashboard

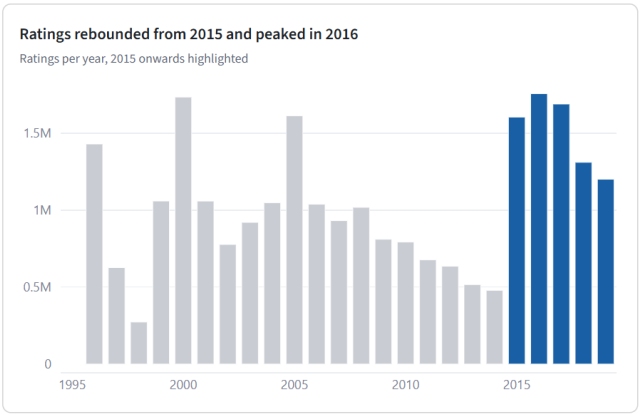
The Streamlit dashboard (dashboard/app.py) reads the Gold and model output Parquet files produced by the Dataproc run. It was designed from business requirements: each page answers the question of one stakeholder group, and every page follows the same reading order of headline numbers, then the main chart, then detail. Chart titles state the finding rather than the metric, one accent colour marks what matters, numbers are abbreviated for readability, and labels use plain language. Run: streamlit run dashboard/app.py

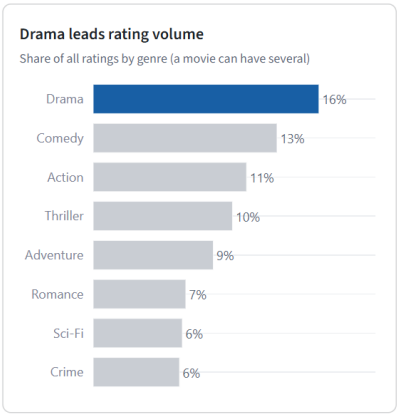
Page	Audience	What it answers
Overview	Leadership	Platform health and recommender impact
Content	Content and acquisition	Top movies per genre, genre volume vs rating, trends, tags, similar movies
Audience	Growth and marketing	Engagement segments, active users over time, rating behaviour
Recommendations	Personalisation	A user's favourites next to their predicted Top-10
Model quality	Data science	RMSE vs baseline, tuning results, error analysis
Data pipeline	Data engineering	Medallion layers, data-quality checks, Dataproc job runtimes

Overview

How the catalogue is used and how well recommendations work

Ratings	Users	Average rating	Prediction error
25.0M	162.5K	3.53	0.79
1995 to 2019	6% are power users	50% of ratings are 4 stars or more	26% better than baseline





Key insights

Audiences are generous. 50% of ratings are 4 stars or more, so a 3-star rating is a negative signal on this platform.

War delights, Horror divides. War averages 3.79 against 3.29 for Horror.

Recommendations are reliable. Predictions land within 0.61 stars on average, 26% better than guessing the average rating.

Figure 6: Dashboard – Overview page

Recommendations

Check a user's taste against what the model suggests

User ID: 148 − + Active user | 132 ratings | average 4.13 | favourite genres: Drama, War, Adventure

Loved

This user's highest rated movies

Title	Genres	Rating
Paths of Glory	Drama, War	5.0
Good, the Bad and the Ugly, The (Buono, il brutto, il cattivo, Il)	Action, Adventure, Western	5.0
Chinatown	Crime, Film-Noir, Mystery, Thriller	5.0
Life Is Beautiful (La Vita è bella)	Comedy, Drama, Romance, War	5.0
Seven Samurai (Shichinin no samurai)	Action, Adventure, Drama	5.0
12 Angry Men	Drama	5.0
Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb	Comedy, War	5.0
Citizenfour	Documentary	5.0
Doctor Zhivago	Drama, Romance, War	5.0
2001: A Space Odyssey	Adventure, Drama, Sci-Fi	5.0

Recommended next

Unseen movies ranked by predicted rating. 1 of 10 share a favourite genre

#	Title	Genres	Predicted
1	Planet Earth II	Documentary	4.54
2	Planet Earth	Documentary	4.53
3	Band of Brothers	Action, Drama, War	4.49
4	In the blue sea, in the white foam.	Animation, Children, Fantasy	4.47
5	Life	Documentary	4.44
6	Cosmos		4.43
7	The Adventures of Sherlock Holmes and Dr. Watson: Bloody Signature	Crime	4.42
8	Alone in the Wilderness	Documentary	4.40
9	Blue Planet II	Documentary	4.31
10	Adventures Of Sherlock Holmes And Dr. Watson: The Twentieth Century Approaches	Crime, Mystery	4.30

Recommendations come from Spark MLlib ALS trained on 25M ratings, limited to movies with 50 or more ratings, and exclude anything the user has already rated.

Figure 7: Dashboard – Recommendations page

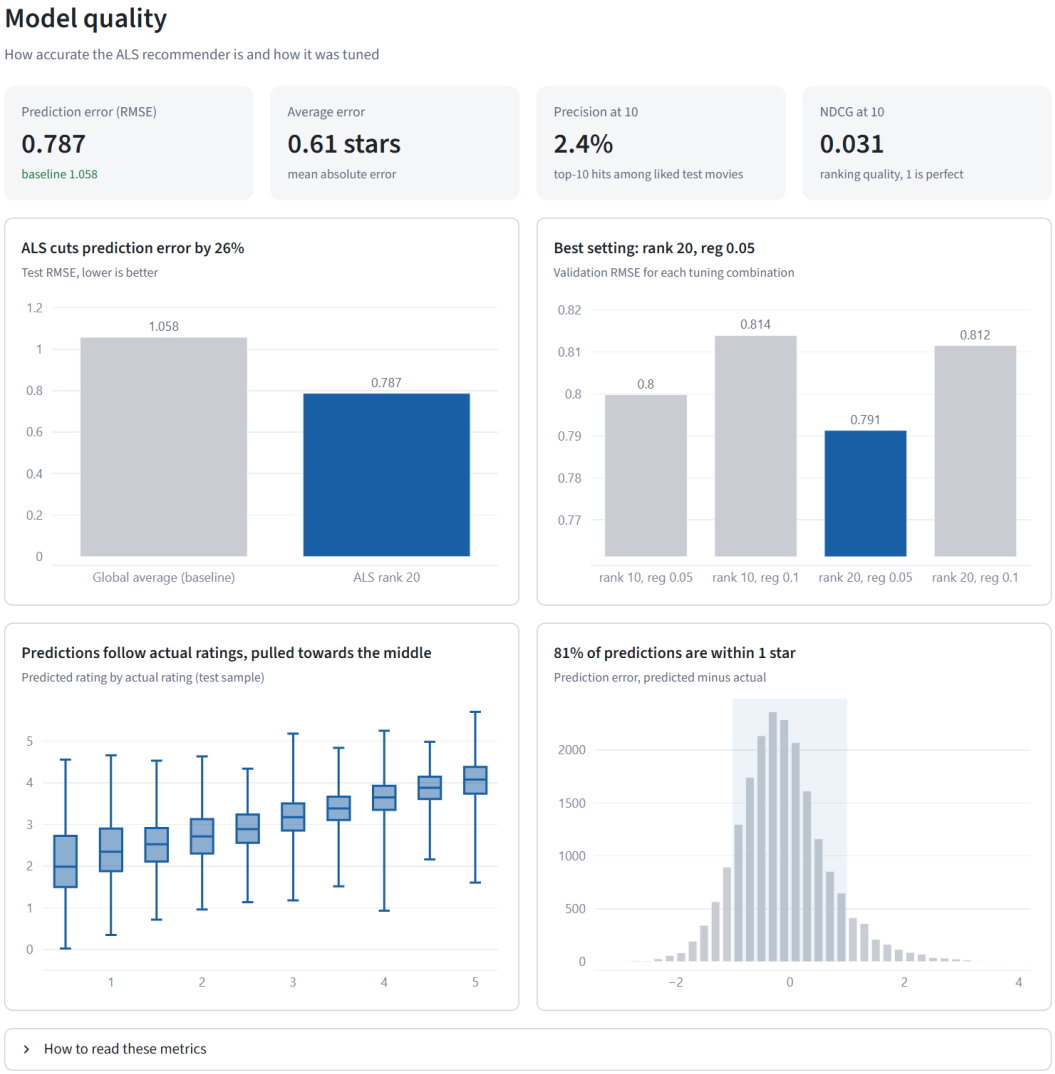


Figure 8: Dashboard – Model quality page

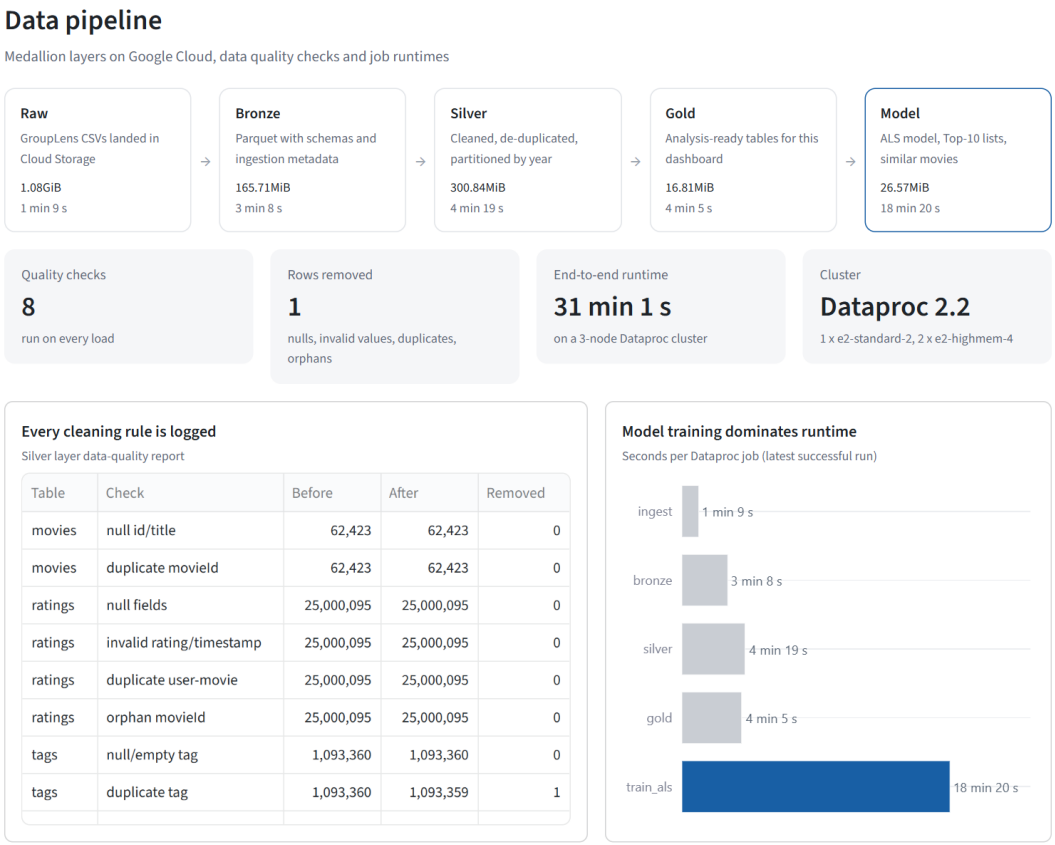


Figure 9: Dashboard – Data pipeline page

7. Cloud execution & orchestration

scripts/dataproc_run.ps1 creates a GCS bucket and a Dataproc 2.2 cluster (auto-deleted after 30 minutes idle), uploads the code, submits ingest, bronze, silver, gold and train_als as PySpark jobs, downloads the outputs for the dashboard and deletes the cluster to minimise cost. The same steps are defined as an Airflow DAG (create_cluster → ingest → bronze → silver → gold → train_als → delete_cluster, weekly schedule, retries, cluster deletion with trigger_rule=ALL_DONE) for deployment on Cloud Composer.

Cloud execution evidence

The repository folder docs/cloud_run_evidence/ contains the Dataproc job history (jobs.txt), the cleaned run logs, the GCS bucket listing per layer (bucket_listing.txt), the Spark driver output of the final ALS job and GCP console screenshots. Job history (project movie-reco-bigdata-2026, region asia-south1):

Job id	Script	State	Start (UTC)	Seconds
d12c51a5	ingest.py	ERROR	2026-10-15 13:47:11	148
177a1681	ingest.py	DONE	2026-10-15 13:50:54	69
e02e75a2	bronze.py	DONE	2026-10-15 13:52:09	188
9cd35a5d	silver.py	DONE	2026-10-15 13:55:27	259
01bdeeb7	gold.py	DONE	2026-10-15 13:59:56	245

0225b782	train_als.py	DONE	2026-10-15 14:04:09	1110
8837fca0	train_als.py	DONE	2026-10-15 14:30:29	1100

GCS layer	**Size**
raw	1.08GiB
bronze	165.71MiB
silver	300.84MiB
gold	16.81MiB
output	26.57MiB
code	18.56kiB

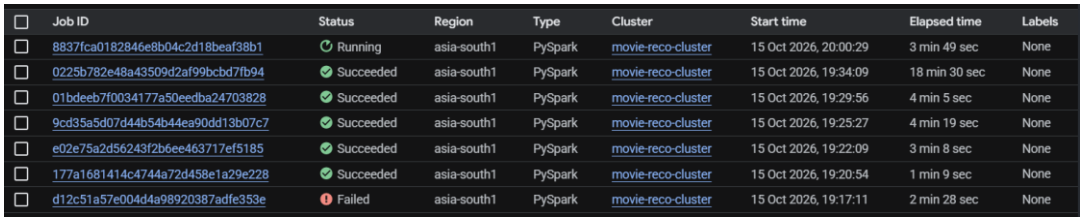


Figure 10: Google Cloud console – Dataproc jobs history

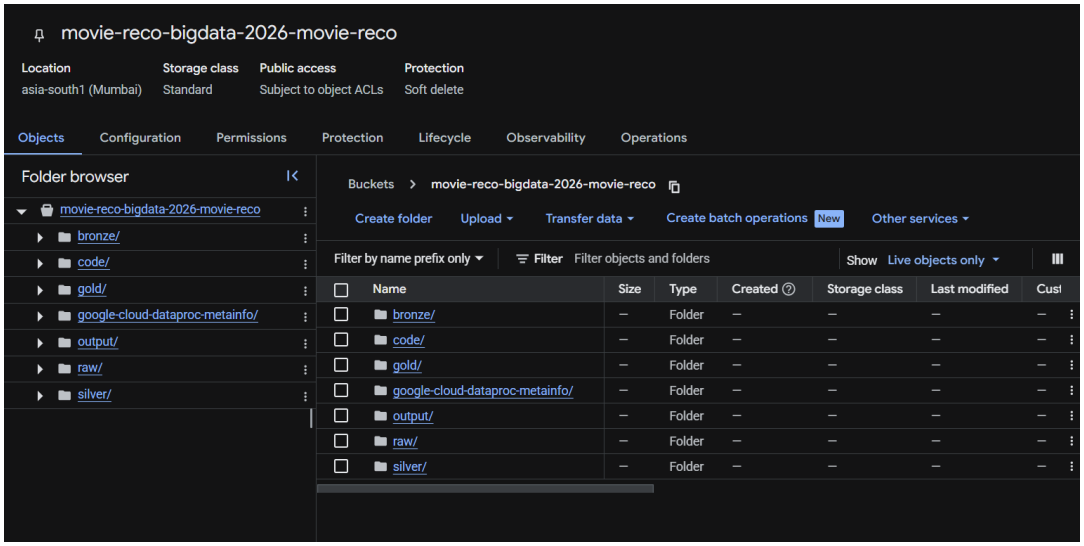


Figure 11: Google Cloud console – GCS bucket medallion layers

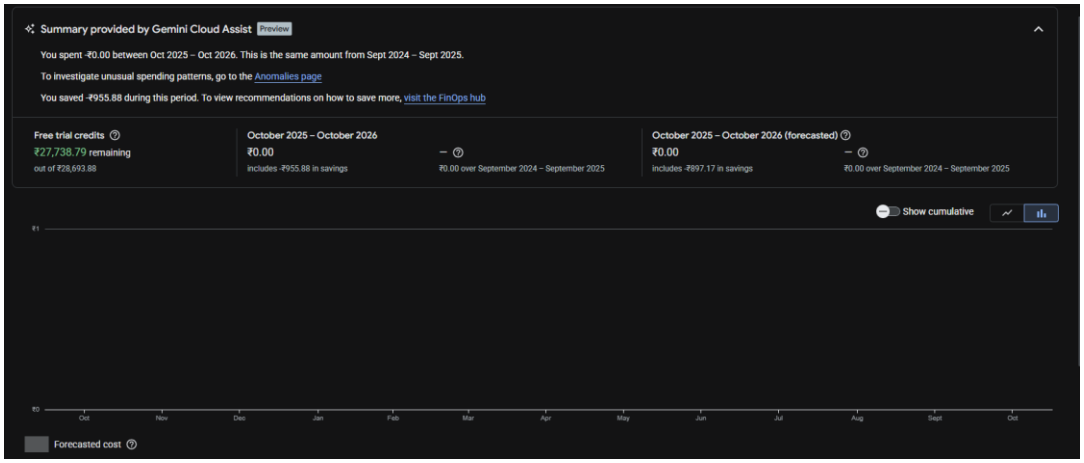


Figure 12: Google Cloud console – Billing free trial credits

Development followed a Git feature-branch workflow: each part (setup/ingest, Bronze/Silver, Gold/ML, cloud orchestration, dashboard/docs) was developed on its own branch and merged through a pull request.

8. Conclusion & Future Work

The project demonstrates scalable Big Data processing of 25,000,095 ratings on Google Cloud: a governed medallion data lake, Spark SQL analytics, a tuned MLlib recommender that clearly beats the baseline, and an interactive dashboard on real outputs. Future work: implicit-feedback ALS and hybrid content features (genres, tags) for cold-start movies, larger hyper-parameter search with CrossValidator, Spark Structured Streaming for real-time rating ingestion, and serving recommendations through an API.

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